LECTURE 1
Introduction to Open Source GIS

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Background

Some definitions

Proprietary Software is software that is **owned by an individual or a company** (usually the one that developed it). There are almost always **major restrictions** on its use, and its source code is almost always kept secret. Examples of proprietary software are Windows, Microsoft Office, MATLAB, ArcGIS, etc.

Open Source Software and/or **Free software** is software where the **source code is made available** under a license that **allows the modification, and re-distribution** of the software at will. Examples of free software/OS systems are Linux, Open Office, R, QGIS, etc.

Freeware, not to be confused with free software, is a type of proprietary software that is **offered for use free of monetary charges**. However, as is the case with other types of proprietary software, there are **generally severe restrictions** on its use and the source code is kept secret. Examples of freeware include Adobe's *Acrobat Reader*.

Background

The concepts of four freedoms of free software (Free Software Foundation)

- The freedom to run the program as you wish, for any purpose (freedom 0).
- The freedom to study how the program works, and change it so it does your computing as you wish (freedom 1). Access to the source code is a precondition for this.
- The freedom to redistribute copies so you can help others (freedom 2).
- The freedom to distribute copies of your modified versions to others (freedom 3). By doing this you can give the whole community a chance to benefit from your changes. Access to source code is a precondition for this.

Background

The most commonly used license, the **GNU General Public License** (GNU GPL), additionally requires that if a modified version of the software is distributed, the source code for such **modified version must be made freely available** (copyleft). The best known example of software licensed under the GPL is the Linux OS.

GNU is an on-going project by the **Free Software Foundation** (https://www.fsf.org/) whose purpose is to create a complete and freely distributable computing environment.

OSI (upcoming slide) lists 64 different OS licenses, 30 of which are compatible with the GNU GPL license.

MIT license is used for many software systems. It allows commercial development of the software (permissive license).

Background

The GNU project was initiated by **Richard Stallman** in 1983 at the Massachusetts Institute of Technology (MIT) when he was attempting to obtain funding for his freely distributable text editor, **GNU Emacs**. Stallman announced his project to the public in newsgroups in September, 1983. Full-scale work began on January 5, 1984, when he resigned from his position at MIT in order to devote his full attention to the project.

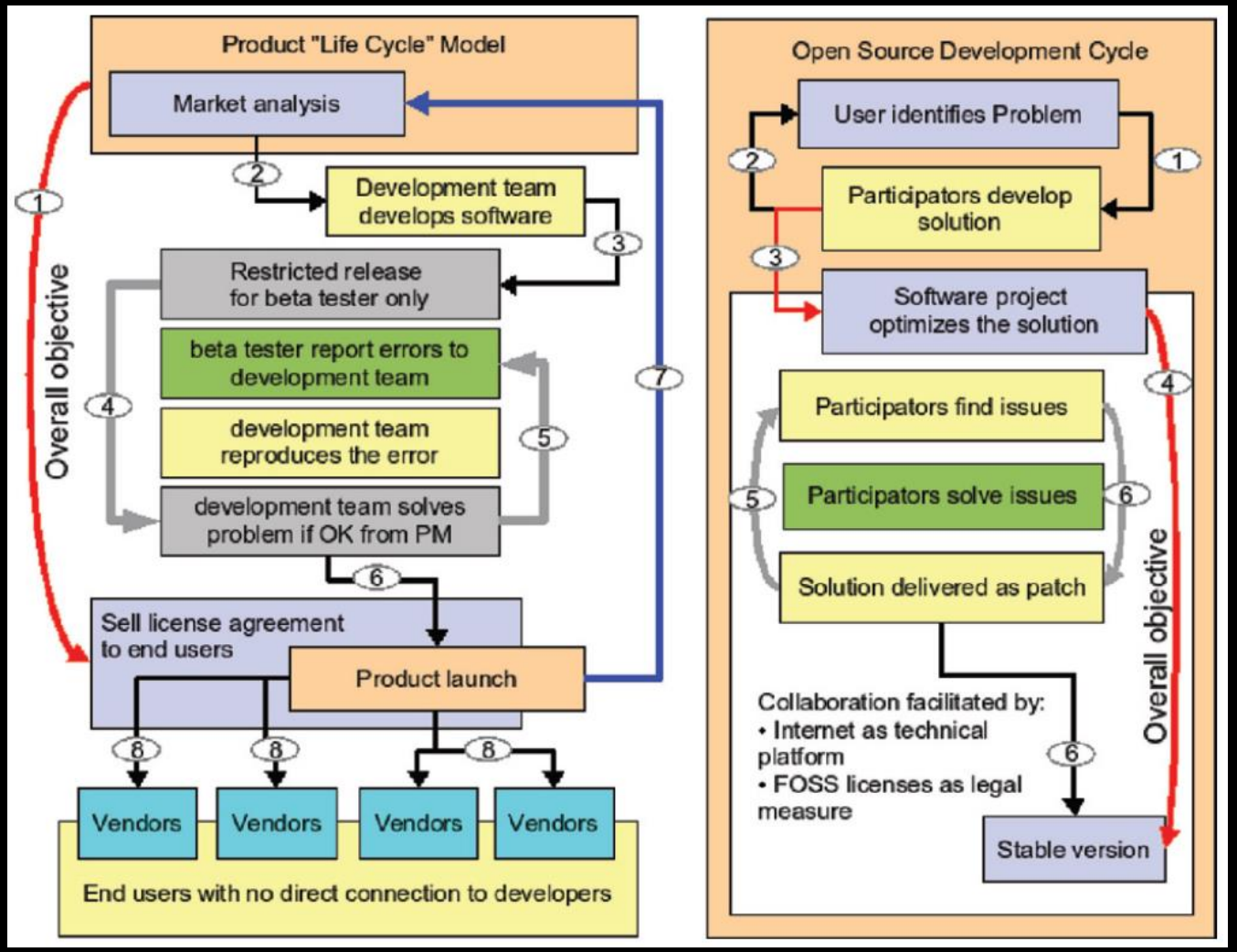
In the 1990s proprietary software companies antagonized the Free and Open-Source Software (FOSS) movement by lobbying and other means (Fear, Uncertainty and Doubt, FUD)

2006 several leading FOSS communities created OSGeo (upcoming slide) in order to build the highest-quality OS geospatial software tools possible.

An incubation process of new software projects is used which will assure high-quality.

Background

Development cycles for proprietary and OS software



Many OS software starts up and dies because no one is using them

Background

www.osgeo.org

The Open Source Geospatial Foundation, or **OSGeo**, (founded 2006) is a not-for-profit organization whose mission is to support and promote the collaborative development of **open geospatial technologies and data**. The foundation provides financial, organizational and legal support to the broader open source geospatial community.

Closely related to *Open Source Initiative* (**OSI**) which is a more general organization for all kind of open source tools, data and software.

https://opensource.org

The membership is composed from users, supporters, promoters, and developers of open source geospatial software. Becoming a member is as easy as joining the site and contributing to the foundation goals.

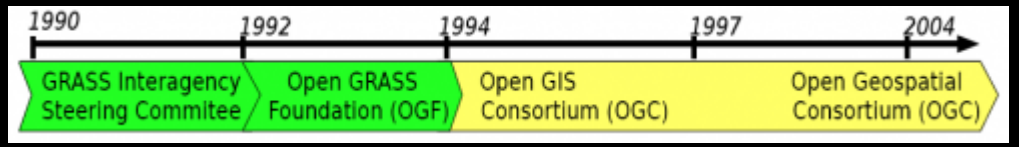
The organization provide resources for foundation projects – e.g. infrastructure, funding, legal and arrange **FOSS4G** conferences.

Background

The **Open Geospatial Consortium**, Inc (OGC) is an international industry consortium of companies, government agencies and universities participating in a consensus process **to develop publicly available interface specifications**. OpenGIS® Specifications support interoperable solutions that "geo-enable" the Web, wireless and location-based services, and mainstream IT. The specifications empower technology developers to make complex spatial information and services accessible and useful with all kinds of applications.

OGC®
Open Geospatial Consortium, Inc.

http://www.opengeospatial.org/



The early history of the OGC is closely related to the Open Source (OS, not to be confused with Operating System) software GRASS.

Because of the non-profit and volunteering mentality within the open source community there **tend to be a lot of branches of software, data etc**. Projects starts and retires. OGC helps to tie everything together and to focus.

Background

Trends in OS GIS (own perspective)

- Desktop OS GIS Software is becoming more user-friendly
- Increased usage in municipalities and agencies
- Many cross platform initiatives (Windows/Linux/Mac OS), e.g. QGIS
- Linux no longer dominating
- Progress in the development of OS Web GIS tools and applications is dominating
- Progress in other areas than just software (e.g. management of spatial databases, spatial data)
- Increasing in number of users and participants

Background

Trends in OS GIS (own perspective)

- Large involvement of commercial companies (e.g. ESRI, and AUTODESK). Make use of LGPL (Lesser General Public License) and MIT-license. ArcGIS Pro list 440 open source software.



- Still most for intermediate or advanced users but rapid development towards beginners

Architecture of most OS Software

A software is usually written in one **computer language** (C++, Java, Python, Ruby, Visual Basic, Perl, …) but usually includes **bindings** to other languages, especially between **C#** and **Python/Perl/Ruby**. The programming language behind an application is important because it affects how the application is distributed, is installed and how easy it is for users to customize.

Most earlier software were based on **command line execution**. **Graphical User Interfaces** (GUI) is now common but still command line execution can be used.

Usually a **main program core with additional plugins/modules**

Architecture of most OS Software

GDAL/OGR

- **GDAL is for raster** and **OGR for vector**. Are used “under the hood” of many GIS applications (GRASS, QGIS, Mapguide, FME, ArcGIS). Supports a huge number of formats.
- All kind of processing and preprocessing tools are included.
- Runs under the MIT license which means that it can be in used in proprietary software (ArcGIS, FME)

PROJ (proj.org)

- Standard library for conversion of geographical coordinates (latitude, longitude) into cartesian coordinates
- Runs under the MIT license which means that it can be in used in proprietary software (ArcGIS, FME)

Open Source desktop GIS

GRASS GIS

GRASS (Geographic Resources Analysis Support System) is the forefather of OSGIS (started 1982) and is now one of the main projects that OSGeo supports. Initially developed by the U.S. Army.

Written in C++ but with bindings to e.g. Python. Runs on different platforms.

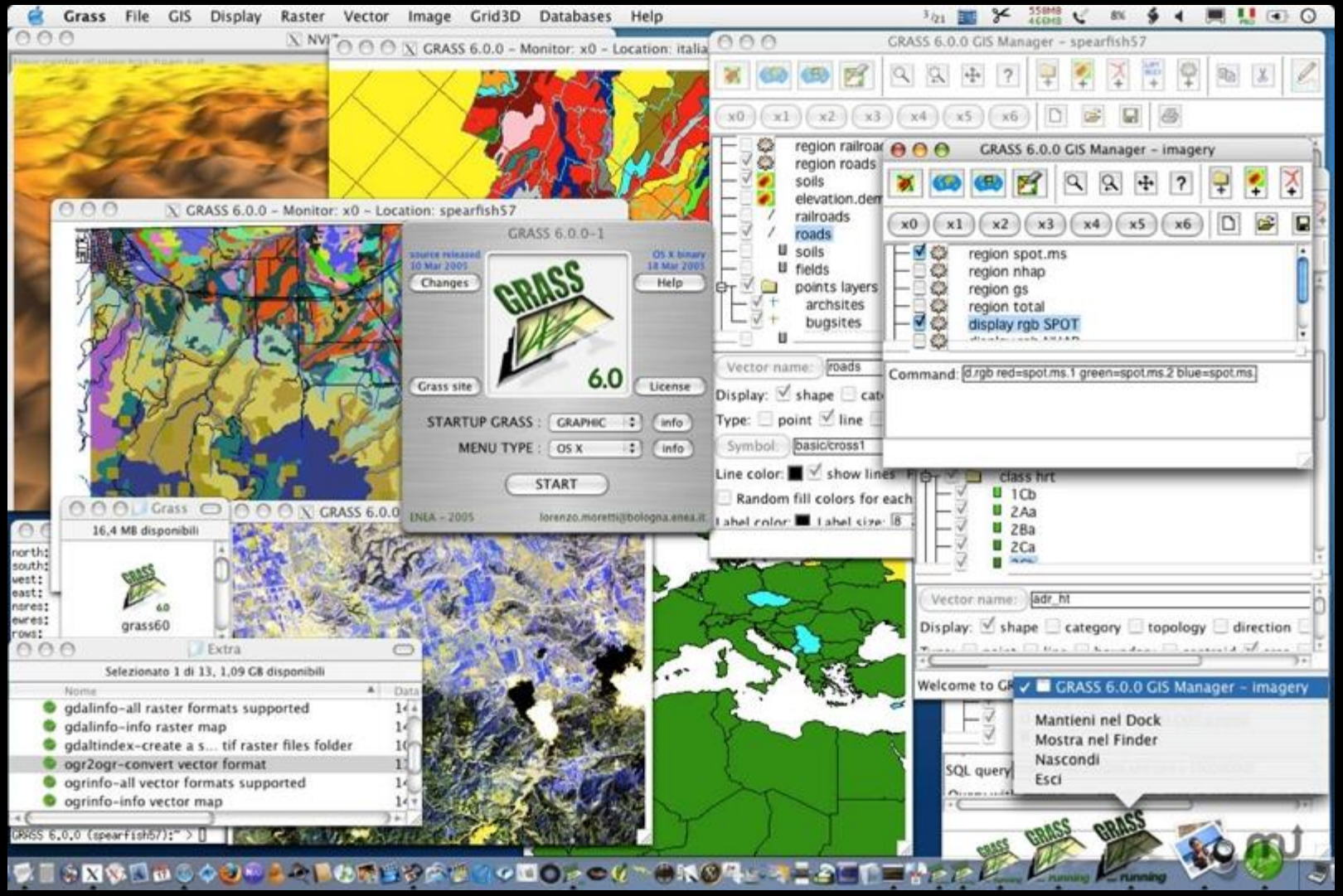
GRASS is both a GUI-program and a command-line program. The real work is done by a suite of command-line programs, or modules.

PROS:
- Stable
- Huge no. of modules (appr. 500 [2026])
- Good documentation
- 3D visualization
- Raster map algebra and simulation tools
- Can be automated and scripted using common scripting languages

CONS:
- Step learning curve
- Own data structure/format
- “Nonstandard” GUI

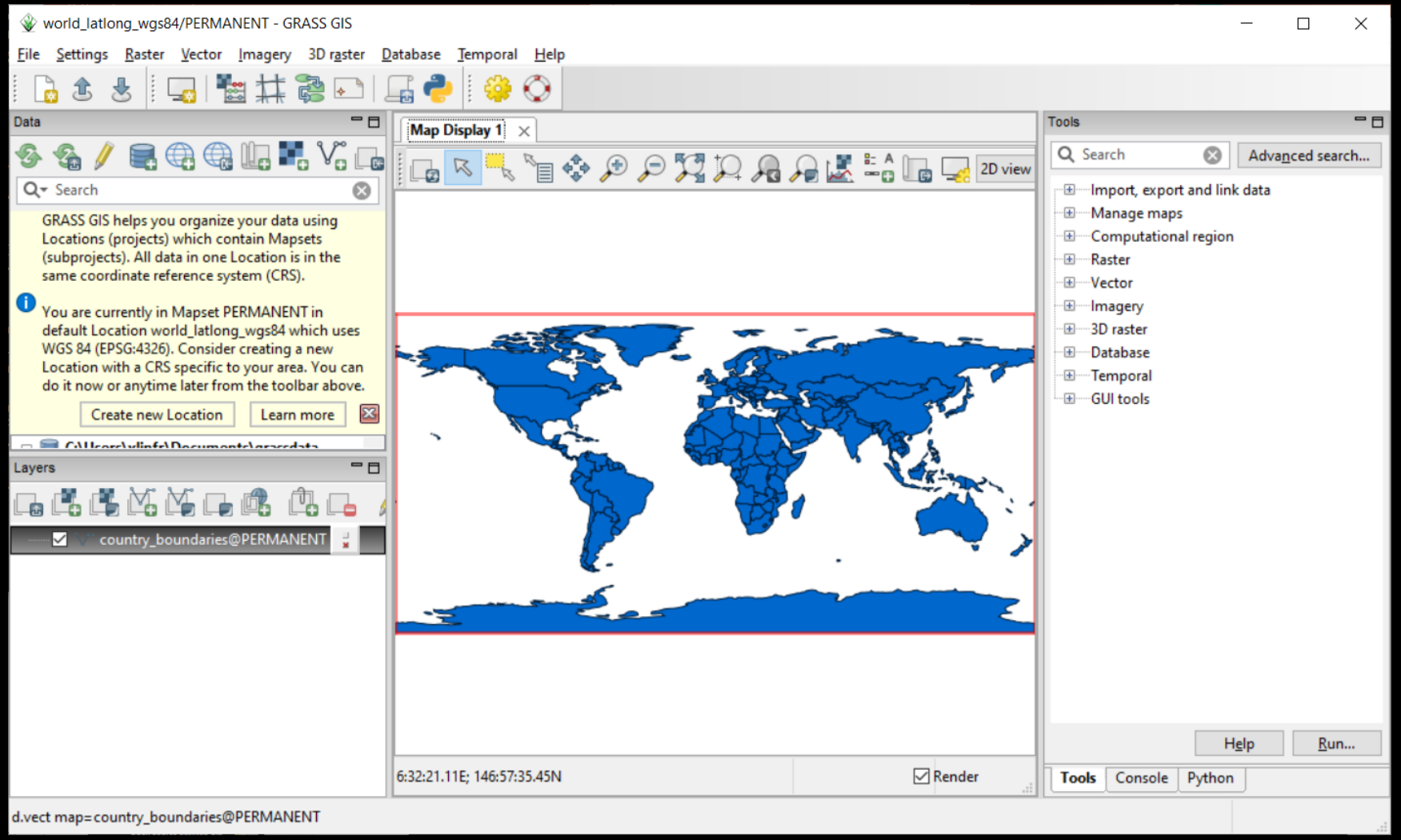
Open Source desktop GIS

GRASS GIS (OLD GUI)



Open Source desktop GIS

GRASS GIS (NEW GUI)



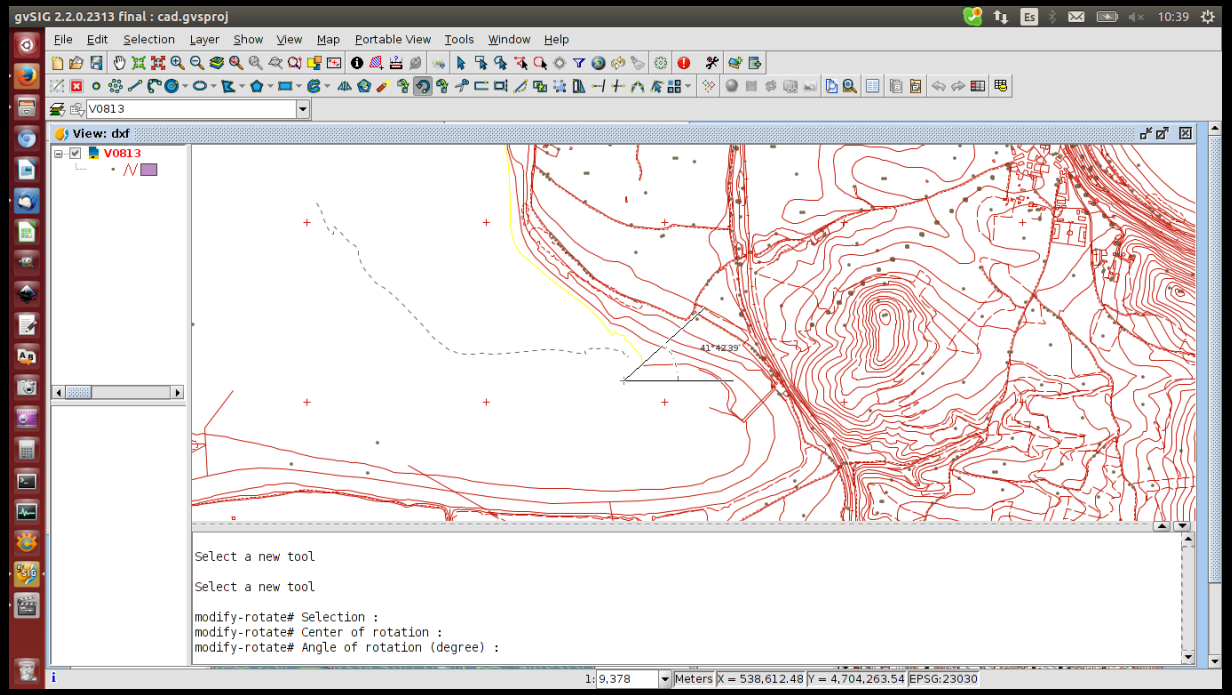
Open Source desktop GIS

gvSIG

A user-friendly OS GIS for both vector and raster. An OSGeo project. Developed in Spain.

Written in Java. And runs on different platforms.

gvSIG is being designed to be easily extendable, allowing continuous application enhancement, as well as enabling the development of tailor-made solutions.



PROS:

- Good format support
- Extensible through plugins
- Map layouts
- Good geoprocessing tools
- Good CAD implementations [2022]

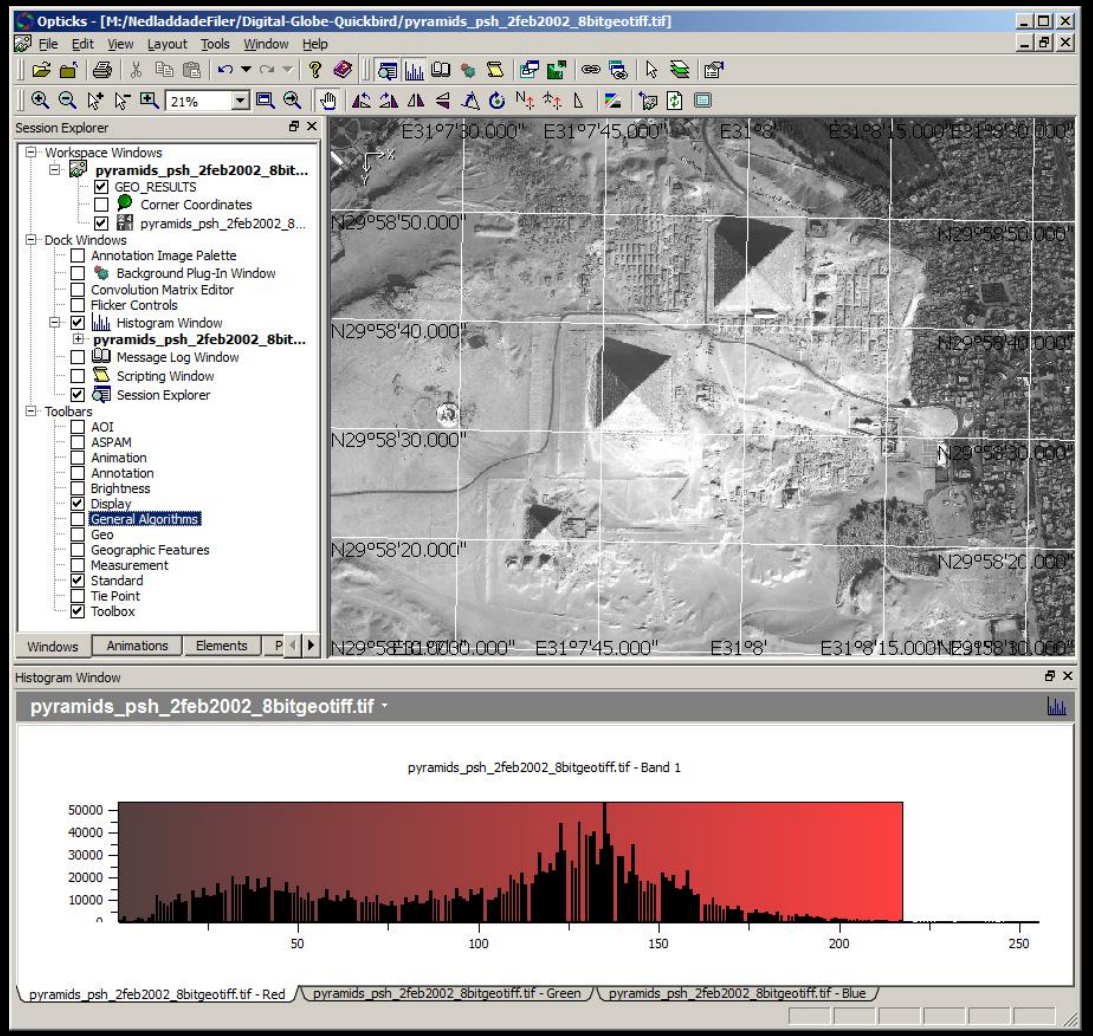
CONS:

- Still incomplete internationalization, i.e. not a lot of documentation in English
- Smaller community support

Open Source desktop GIS

Opticks

Opticks is an expandable remote sensing and imagery analysis software platform. Originally developed by Ball Aerospace for the US Air Force.



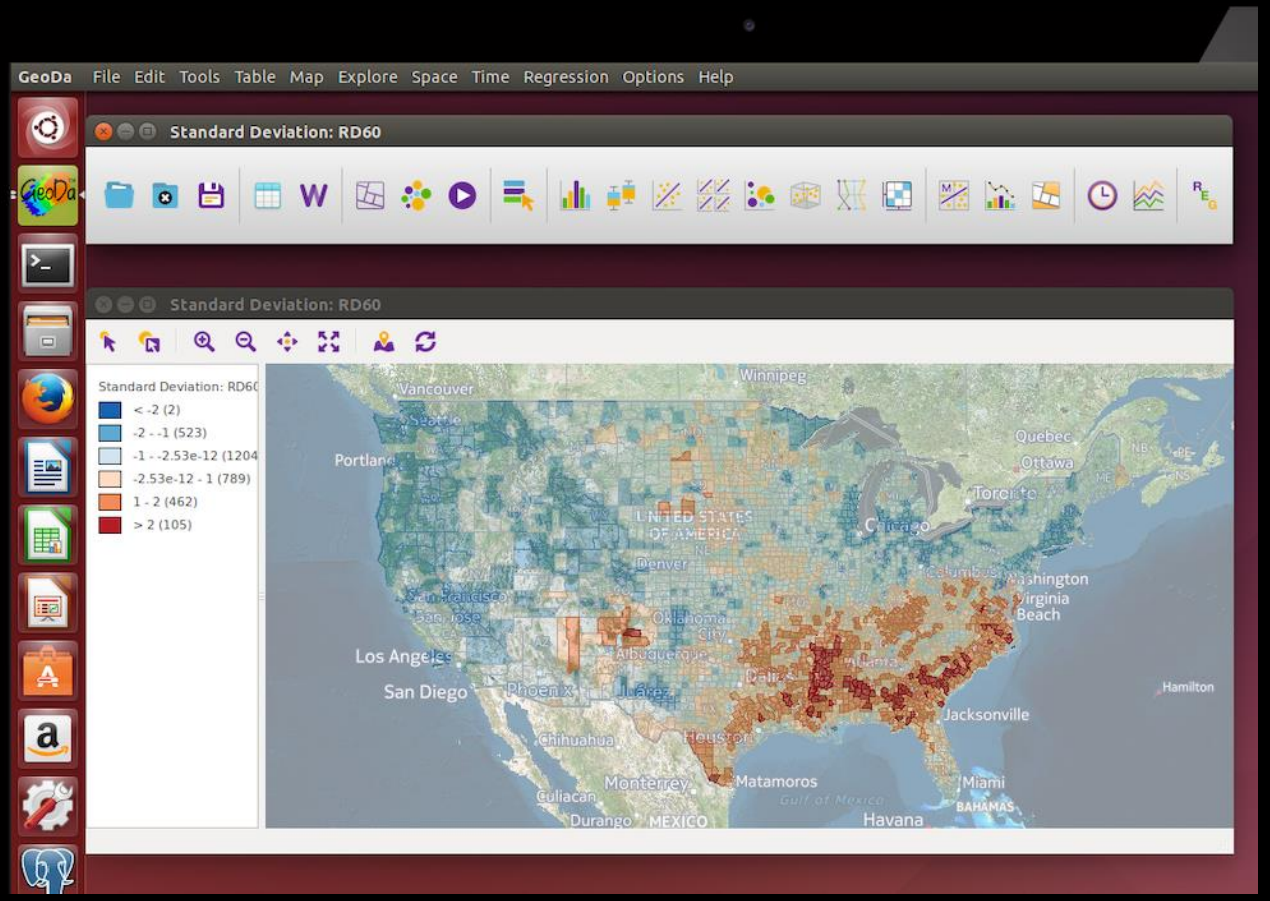
PROS and CONS:

- Not very active community
- Free and open source
- Latest version from 2021(?) according to their repository

Open Source desktop GIS

GeoDa

GeoDa is a free and open source software tool that serves as an introduction to spatial data science. It is designed to facilitate new insights from data analysis by exploring and modeling spatial patterns.



PROS:

– Specialized in statistics

CONS:

– Missing traditional geoprocessing tools

Open Source desktop GIS

SAGA GIS

Stands for **S**ystem for **A**utomated **G**eoscientific **A**nalyses. Originates from the department of Physical Geography in Göttingen University. Written in C++. Based on modules

Available through new plugin in QGIS (SAGA Next Gen)



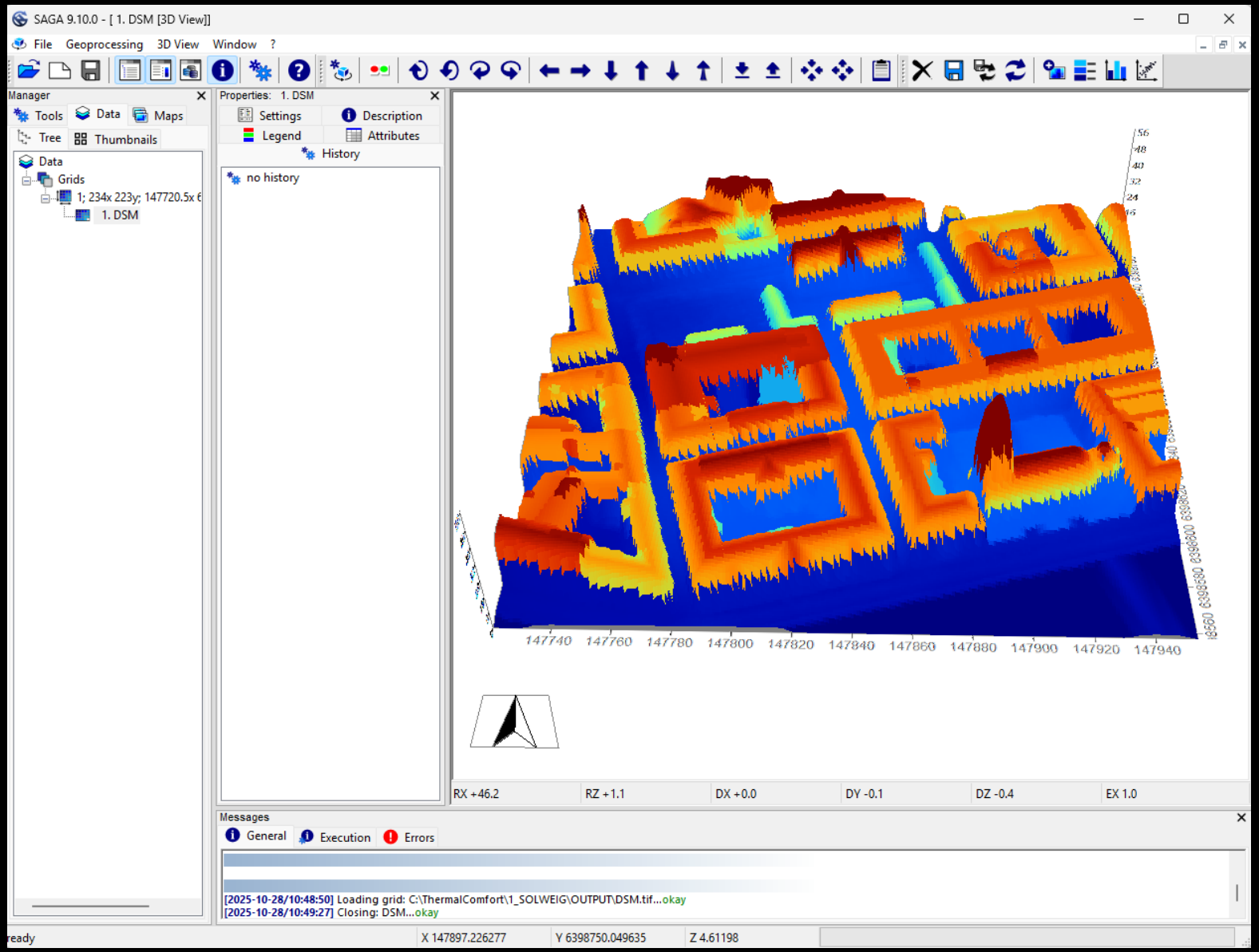
PROS:

- Specializing on geoscience
- Easy and fast
- 3D capabilities

CONS:

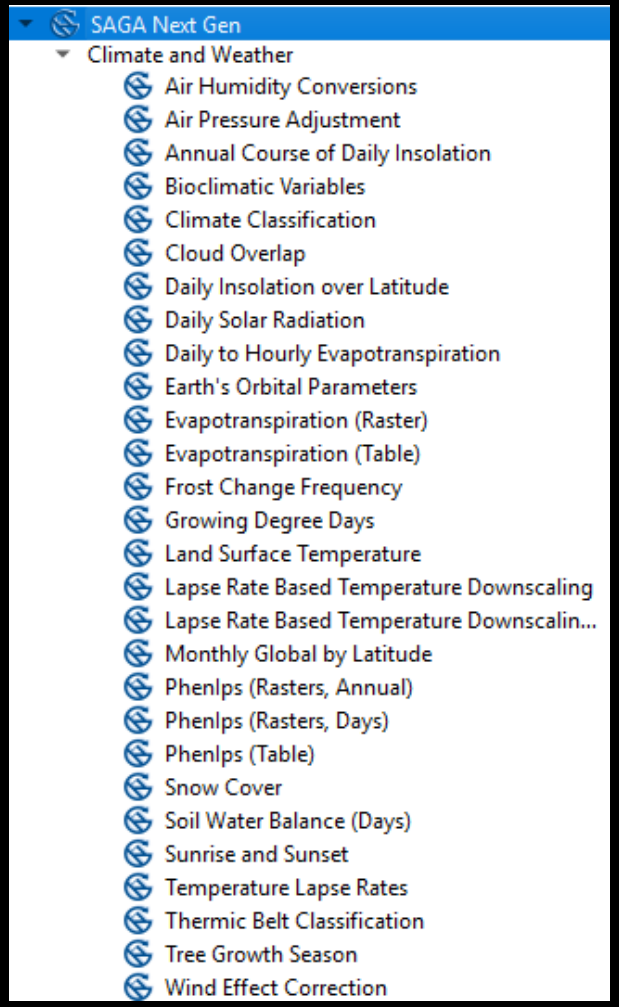
- Missing core documentation for many tools?

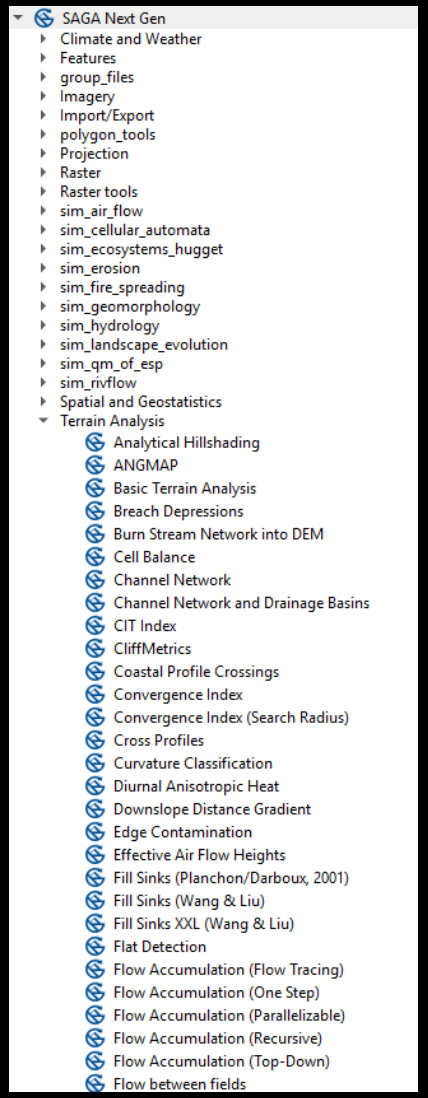
Open Source desktop GIS



Open Source desktop GIS

SAGA Next Gen (QGIS)

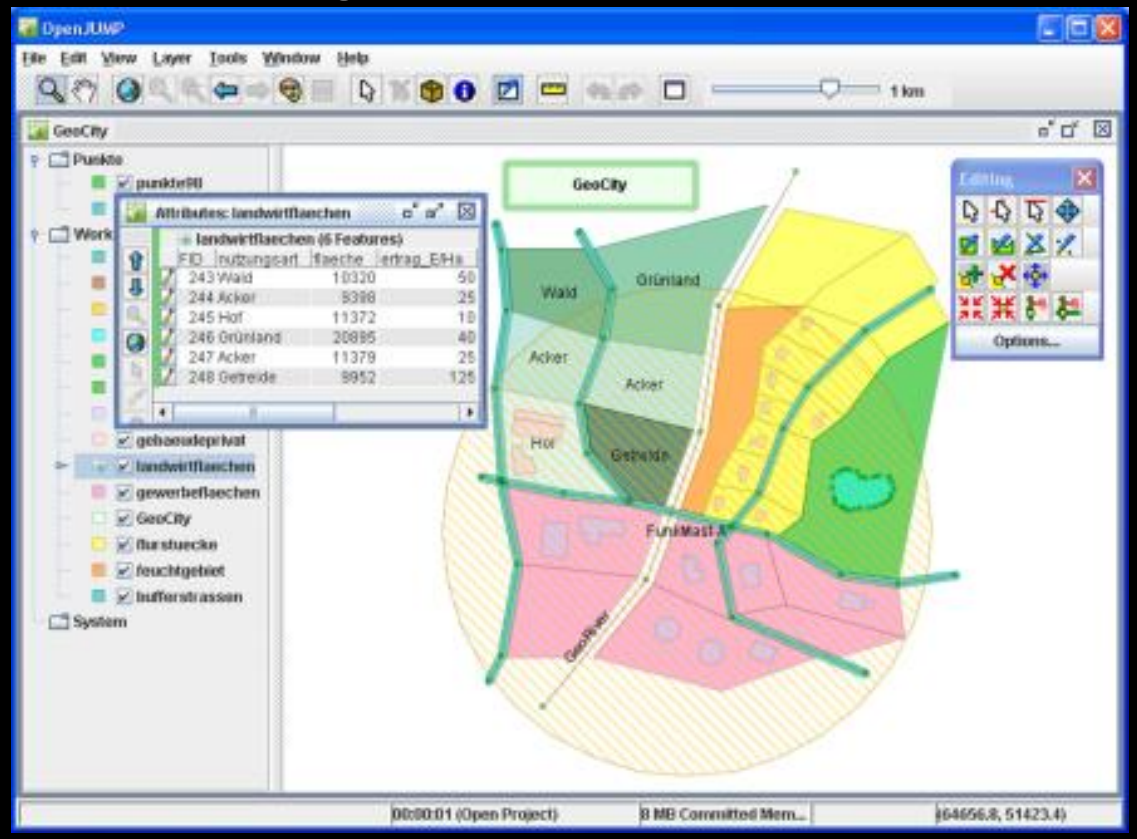




Open Source desktop GIS

OpenJUMP

OpenJUMP is an open source Geographic Information System (GIS) written in the Java programming language. OpenJUMP started as JUMP GIS designed by Vivid Solutions. Both program are similar but deviates sometime. A fork of versions exists (i.e. many versions of OpenJUMP)



PROS:

- Easy install
- Editing tools
- Many formats
- WMS support
- Geometry validation

CONS:

- Six versions available which to choose from with different levels of support, goals and features. Example of a classic setback in OSGIS.

Open Source desktop GIS

OSSIM

Focus is on **image processing and raster processing**. Open Source Software Image Map (OSSIM) is a high performance engine for remote sensing, image processing, geographical information systems and photogrammetry. It has been actively developed since 1996. Pronounced as “awesome”. Was supported by OSGeo.

Has a globe-style viewer, ossimPlanet (like Google Earth and ArcGlobe).

PROS:
- Impressive raster processing capabilities
- Huge amount of data linked to ossimPlanet
- Many formats and platforms

CONS:
- Steep learning curve
- Not well documented

Open Source desktop GIS

Marble

Marble is a virtual globe and world atlas — your swiss army knife for maps.
Has a globe-style viewer, ossimPlanet (like Google Earth and ArcGlobe).





PROS:

– Available for mobile units (2020?) as well as desktop

CONS:

– Some geodata is not high-resolution

Open Source desktop GIS

QGIS (previously Quantum GIS)

Founded in 2002. QGIS is an official project of the OSGeo.

QGIS provides a continuously growing number of capabilities provided by core functions and plugins. You can visualize, manage, edit, analyze data, and compose printable maps.

Written in C++ and binds with Python. Parts of the GUI is written in Python. Runs on multiple OS platforms. Make use of GDAL/OGR libraries (upcoming slide). Can also handle LiDAR data through PDAL.

PROS:

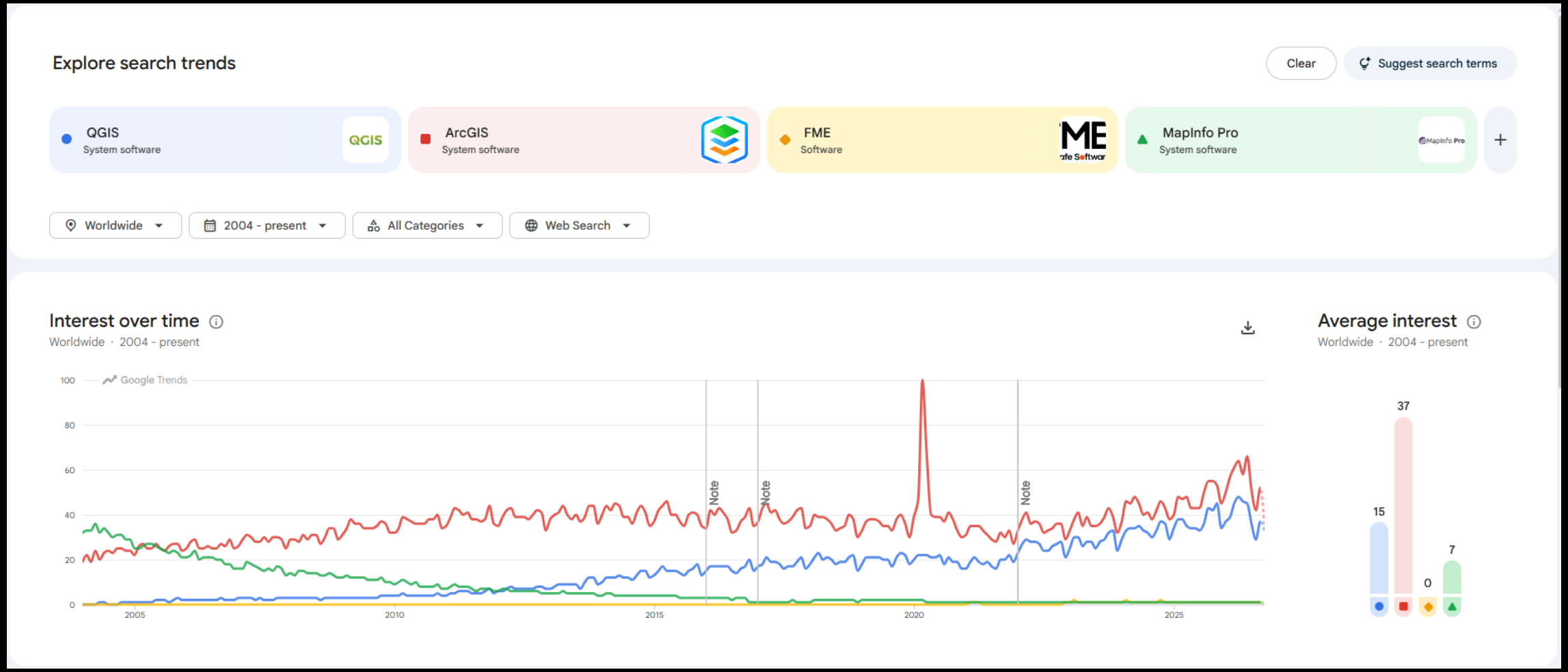
- Supports a wide range of data formats
- Good documentation
- Strong community support (many users)
- Extensible through plugins
- Good integration with GRASS
- Customization using Python

CONS:

- Labeling
- Limited map composition and printing capability
- Still maturing
- 3D, still developing

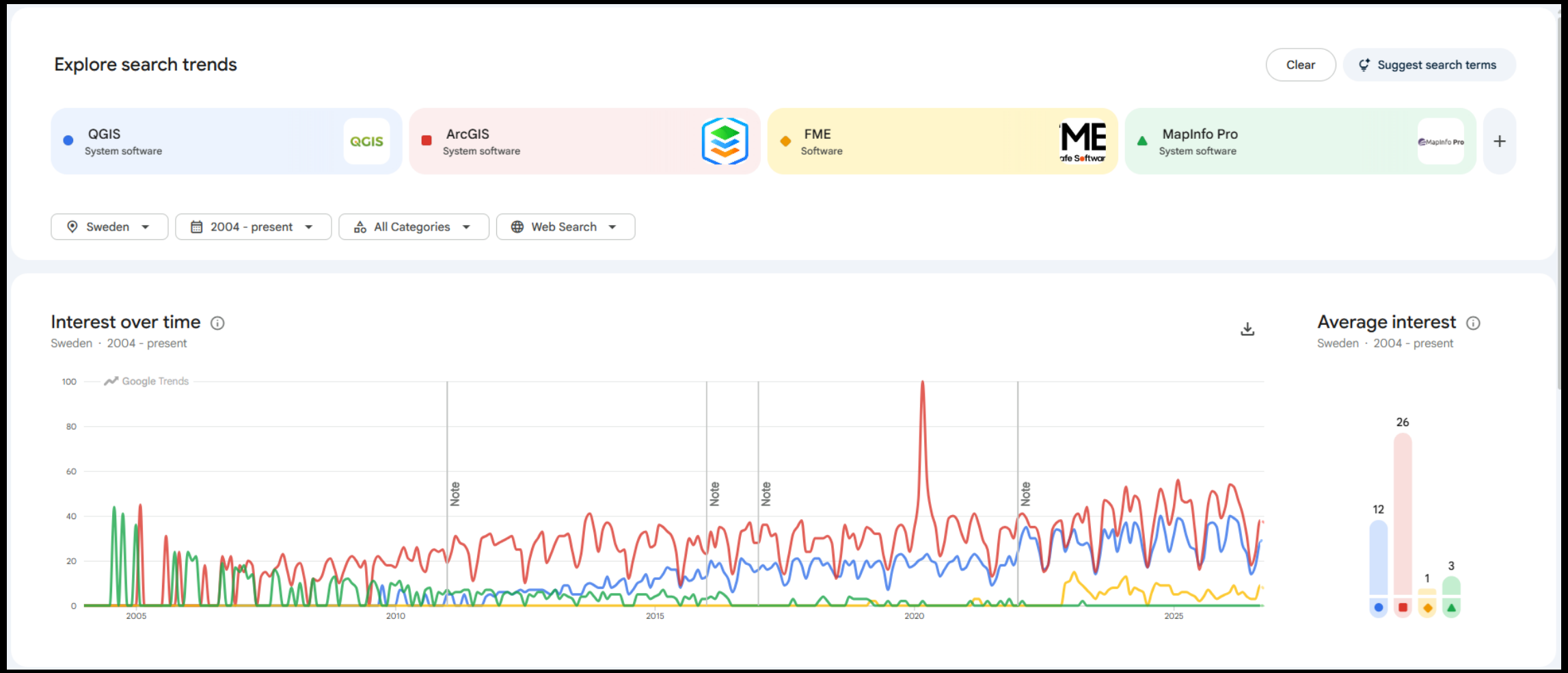
Desktop software development for GIS

Data on search volume since 2004 (trends.google.com)

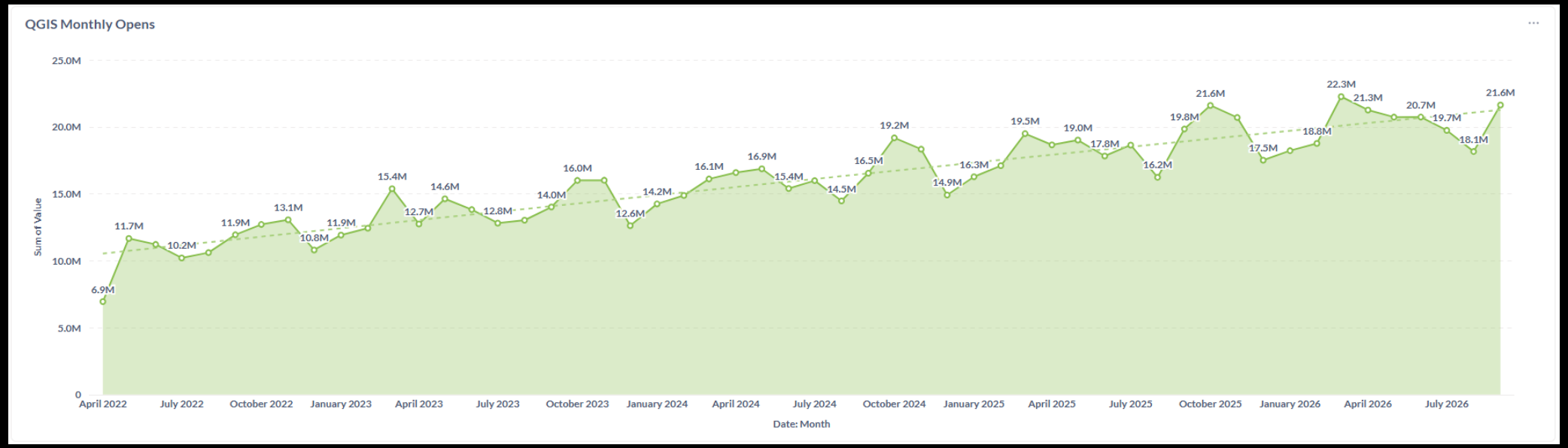


Desktop software development for GIS

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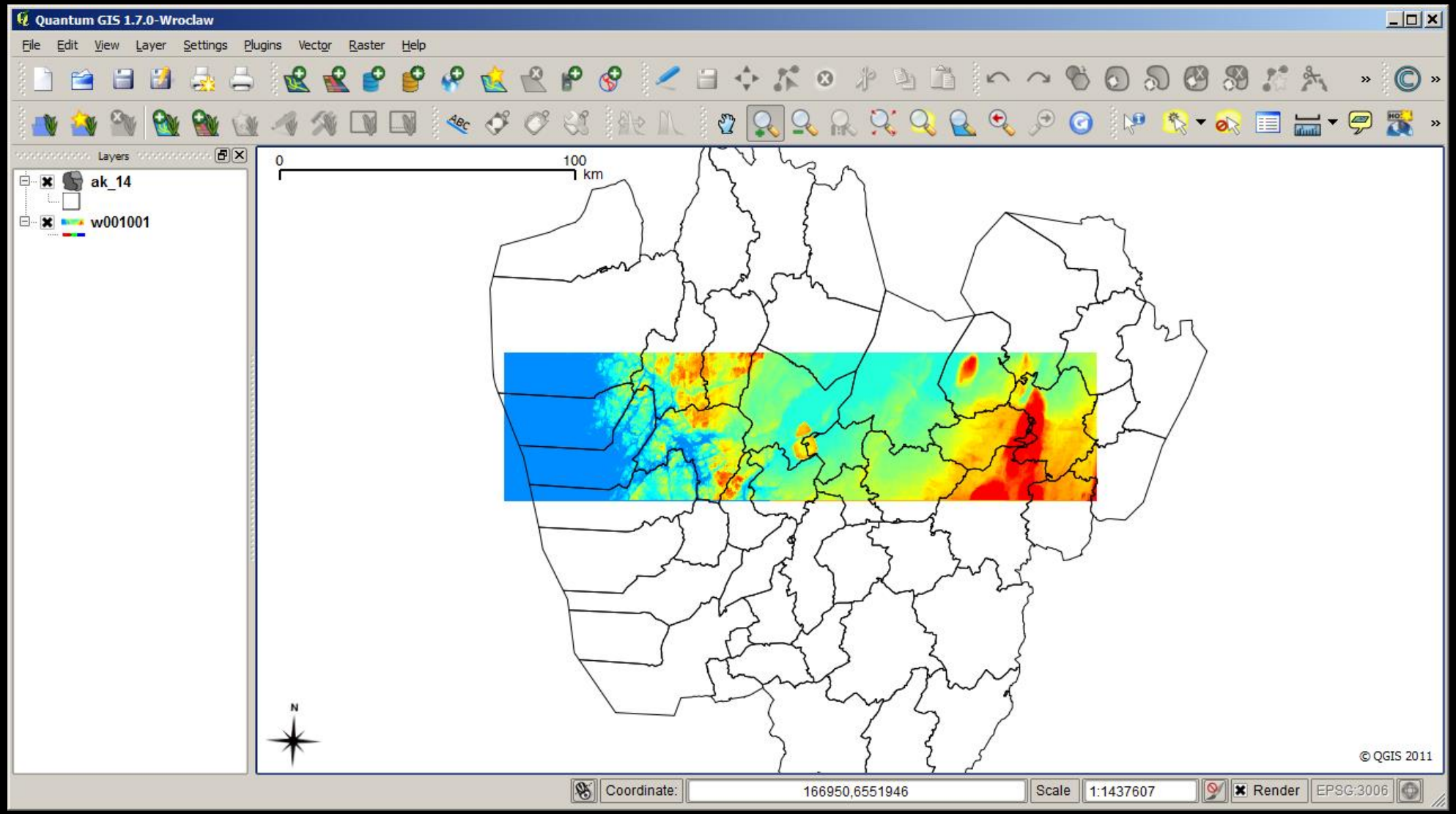


Open Source desktop GIS

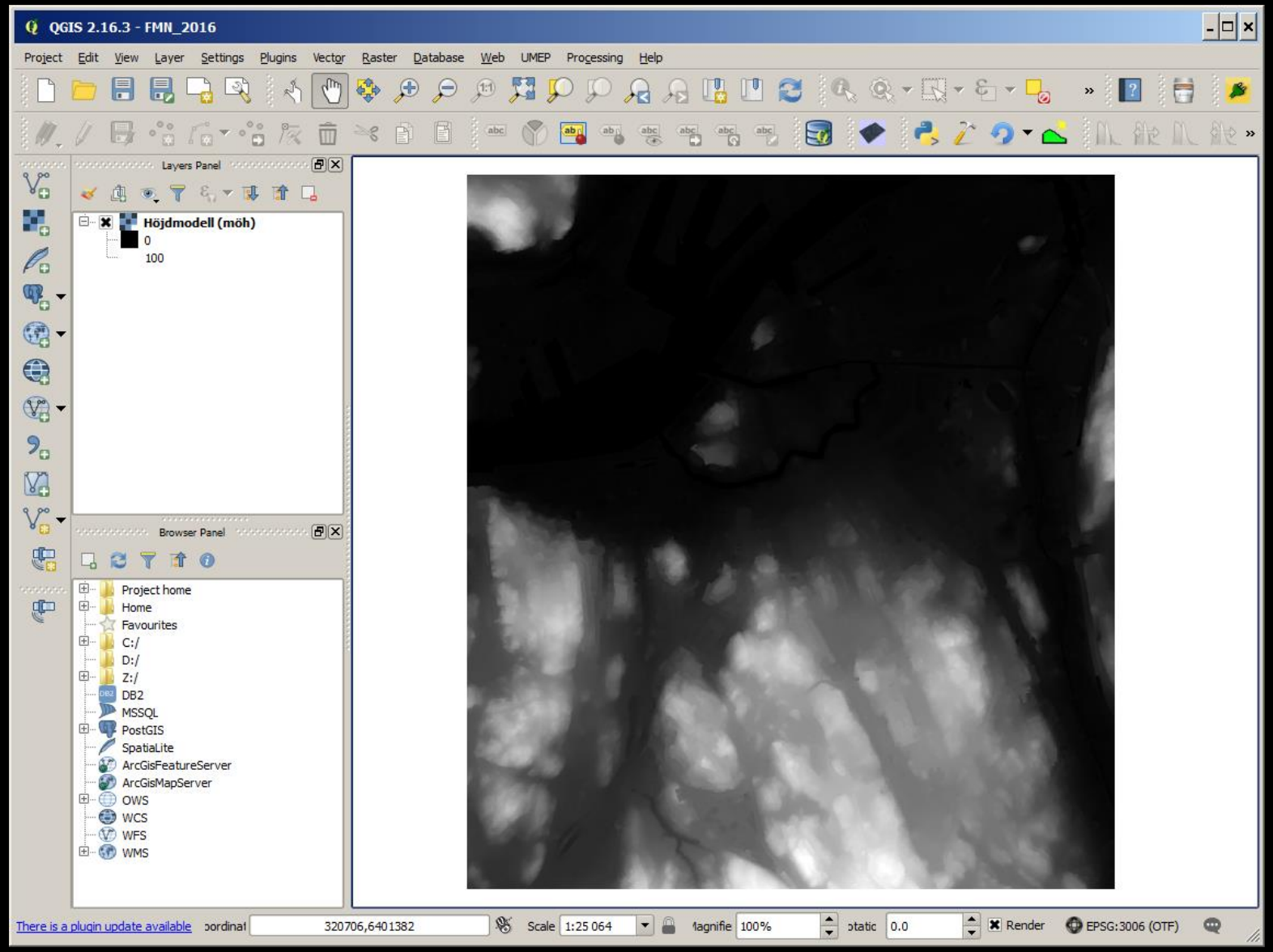




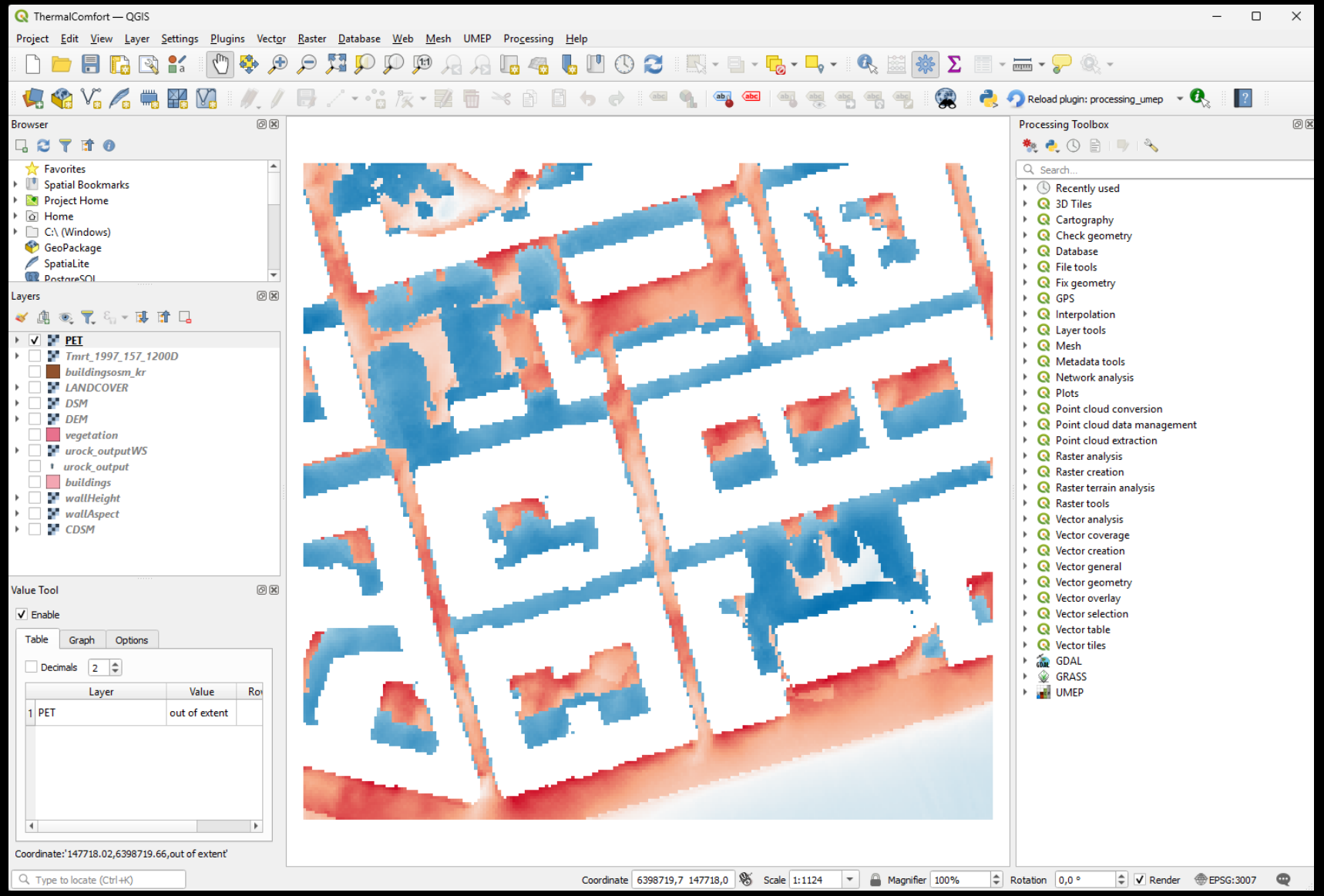
Open Source desktop GIS QGIS



Open Source desktop GIS



Open Source desktop GIS QGIS



QGIS Sweden

Founded in 2013, in Gothenburg

Promotes the use of QGIS, and development of plugins

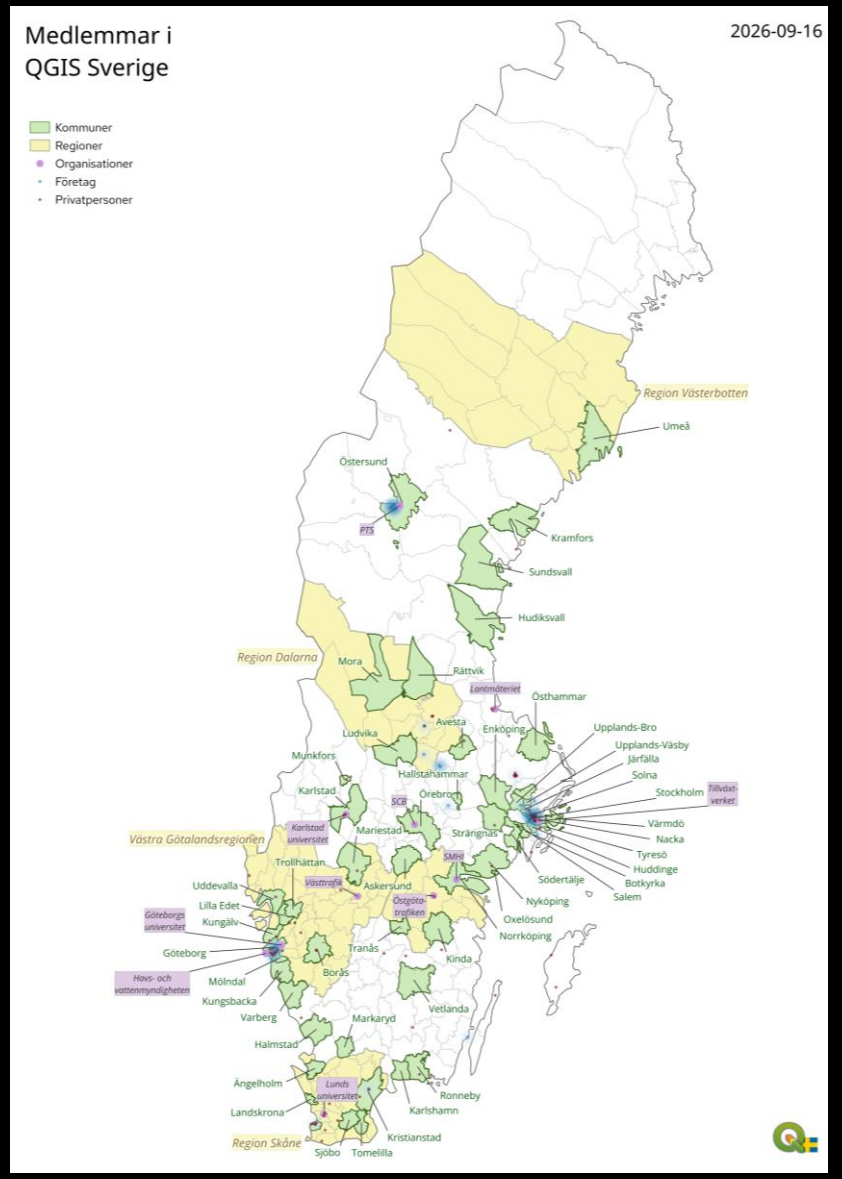
- FUSION (LiDAR data)
- Lantmäteriet data

One in-person and one digital meeting every year

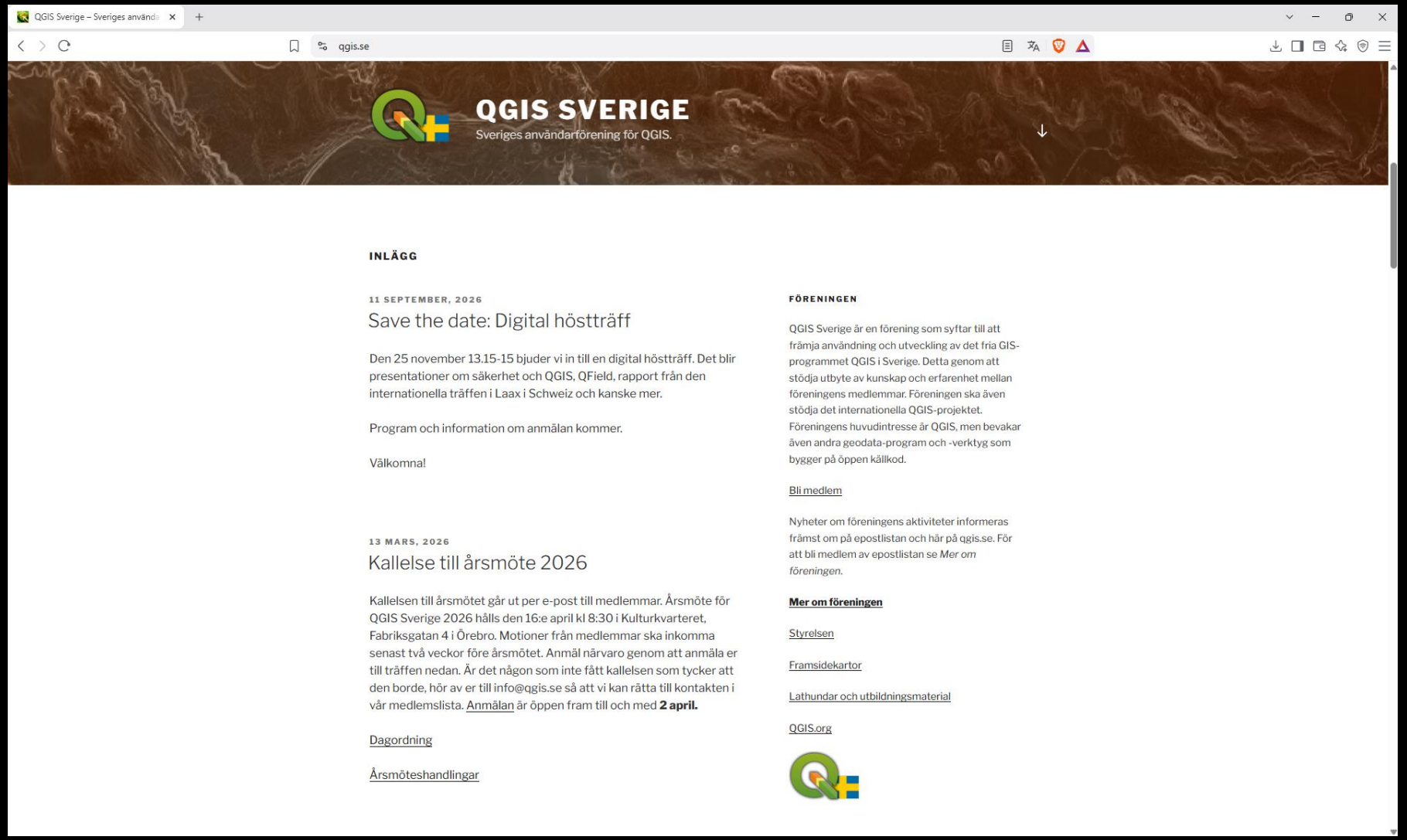
- Digital meeting 25th of November

Around 100 organizations as members

QGIS Sweden is a Flagship member to the QGIS project (QGIS.org)



QGIS Sweden


QGIS Sverige – Sveriges använda
qgis.se
QGIS SVERIGE
Sveriges användarförening för QGIS.
INLÄGG
11 SEPTEMBER, 2026
Save the date: Digital höstträff
Den 25 november 13.15-15 bjuder vi in till en digital höstträff. Det blir presentationer om säkerhet och QGIS, QField, rapport från den internationella träffen i Laax i Schweiz och kanske mer.
Program och information om anmälan kommer.
Välkomna!
13 MARS, 2026
Kallelse till årsmöte 2026
Kallelsen till årsmötet går ut per e-post till medlemmar. Årsmöte för QGIS Sverige 2026 hålls den 16:e april kl 8:30 i Kulturkvarteret, Fabriksgatan 4 i Örebro. Motioner från medlemmar ska inkomma senast två veckor före årsmötet. Anmäl närvaro genom att anmäla er till träffen nedan. Är det någon som inte fått kallelsen som tycker att den borde, hör av er till info@qgis.se så att vi kan rätta till kontakten i vår medlemslista. Anmälan är öppen fram till och med 2 april.
Dagordning
Årsmöteshandlingar
FÖRENINGEN
QGIS Sverige är en förening som syftar till att främja användning och utveckling av det fria GIS-programmet QGIS i Sverige. Detta genom att stödja utbyte av kunskap och erfarenhet mellan föreningens medlemmar. Föreningen ska även stödja det internationella QGIS-projektet. Föreningens huvudintresse är QGIS, men bevakar även andra geodata-program och -verktyg som bygger på öppen källkod.
Bli medlem
Nyheter om föreningens aktiviteter informeras främst om på epostlistan och här på qgis.se. För att bli medlem av epostlistan se Mer om föreningen.
Mer om föreningen
Styrelsen
Framsidekartor
Lathundar och utbildningsmaterial
QGIS.org

QGIS Sweden

User Conference 2025 in Norrköping!



QGIS Sweden



QGIS Sweden



List of OS GIS Software

https://gisgeography.com/

OS command-line applications

GDAL/OGR

Using command-line execution is **usually much more effective** and take a fraction of time compared to GUI-based applications.

Runs under the MIT license (LGPL) which means that it can be in used in proprietary software

GDAL is for raster and **OGR for vector**. Are used “under the hood” of many GIS applications (GRASS, QGIS, Mapguide, FME, ArcGIS). Supports a huge number of formats.

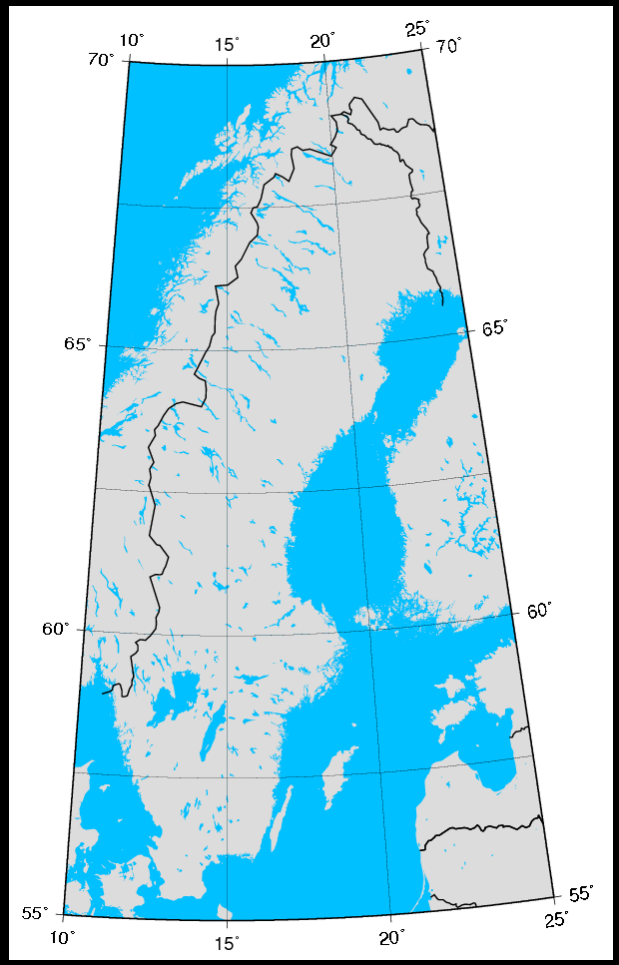
All kind of processing and preprocessing tools are included.

Command-line example:
gdalinfo c:\temp\rasterdata.asc

OS command-line applications

GMT

GMT stands for **the Generic Mapping Tools**. It is also a command-line software which could be used to, among other things, generate overview maps and manipulating geographic and Cartesian data sets. It has a collection of about 65 tools. Some datasets are included which could be used to make maps.



Command-line example:

```
pscoast -JU33N/13c -R10/25/55/70 -G220/220/220 -S0/192/255 -Bxa5g5 -Bya5g2.5
-B+gwhite -Di -P -N1/1p,black > bild.eps
```

Source code repositories

Is a file archive and web hosting facility for source code of software, documentation, web pages, and other works, accessible either publicly or privately. They are often used by open-source software projects and other multi-developer projects to maintain revision and version history, or version control. Many repositories provide a bug tracking system, and offer release management, mailing lists, and wiki-based project documentation.

Popular repositories are GitHub, BitBucket, SourceForge, GitLab

QGIS on GitHub: https://github.com/qgis/QGIS

Integrated Development Environment (IDE) for programming and scripting

A software application that provides comprehensive facilities to computer programmers for software development. An IDE normally consists of at least a source code editor, build automation tools and a debugger.

Build automation tools = Translating code into machine language etc.
Debugger = Run your code under controlled conditions

Popular IDEs for Python are:
GNU Emacs (GPL v3, strong copyleft)
PyCharm (Apache 2.0, permissive)
VSCode (MIT for its source code, permissive)
Jupyter Notebook (Modified BSD, permissive)
Spyder (MIT, permissive)
Eclipse (EPL 2.0, weak copyleft)

List of Python IDEs:
https://en.wikipedia.org/wiki/Comparison_of_integrated_development_environments#Python

LECTURE 2

Open Source GIS Data
Sources and Formats

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Data formats

Open Source (OS) software

Open source software is software whose source code is publicly accessible and can be used, modified, and redistributed under an open source license.

Open source development encourages collaboration, transparency, and the exchange of ideas, allowing communities to contribute to technological and scientific advancement.

Different licenses impose different conditions:

- MIT: Permits modification and redistribution, including as part of proprietary commercial software.
- GNU GPL: Allows modification and commercial use, but redistributed derivative works must generally remain available under the GPL, with corresponding source code.

QGIS is an example of open-source GIS software, distributed under the GNU GPL version 2 or later.

Data formats

OS data formats

An open data format is a format whose technical specifications are publicly available and can be implemented without restrictive licensing conditions.

Open formats promote interoperability between different software applications and reduce dependence on individual software vendors.

Examples: CSV, PNG, JSON, XML, GeoJSON, GeoPackage, and GML.

Open GIS data formats

GIS software often supports a wide range of geospatial formats through libraries such as GDAL/OGR.

GDAL provides support for both open and proprietary formats, making data exchange between different GIS applications easier.

However, not every format supports the same operations, and some can only be read or have limited writing capabilities.

Data formats

The need for standards

The large number of GIS data formats and software applications creates challenges when exchanging and sharing geospatial information.

Open standards provide common specifications for how geographic information should be structured, stored, and exchanged.

This promotes interoperability, allowing different applications to work with the same data without depending on a single software vendor.

The Open Geospatial Consortium (OGC) develops open standards for geospatial information, including GeoPackage, GML, WMS, and WFS.

Other standardization organizations, such as ISO (CRS) and IETF (e.g. GeoJSON), also contribute to geospatial standards.

Example of the GeoPackage standard: https://www.ogc.org/standards/geopackage/

Data formats

Open formats vs. Open data

Open data formats describes how data is structured and stored

Openness is determined by technical specifications and implementation rights

Examples: GeoJSON, GeoPackage, GeoTIFF

Open data describes who can access, use, modify, and redistribute data

Openness is determined by licensing and conditions of use

- CC0 – Free use, modification and redistribution, including commercial use. No attribution required.
- CC BY 4.0 – Same freedoms, but attribution is required.
- ODbL 1.0 – Attribution required. Publicly used adapted databases must also be made available under ODbL.

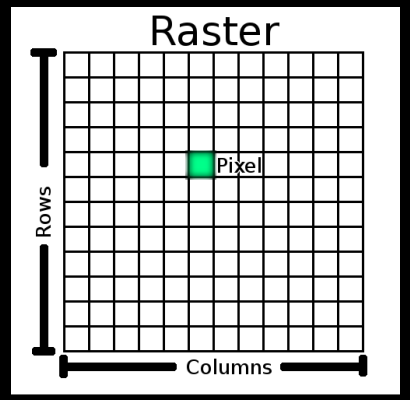
Examples: OpenStreetMap, openly licensed government geodata

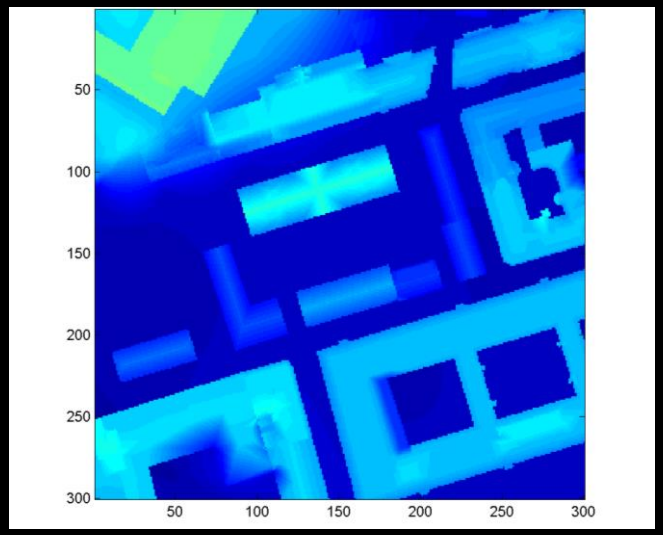
Data formats

Repetition: Raster vs. Vector

Raster, e.g. .tif

- Grid of cells (pixels)
- One value per pixel per band
- Can have multiple bands (e.g. RGB = 3 values/pixel)
- Example: DEM (Digital Elevation Model)

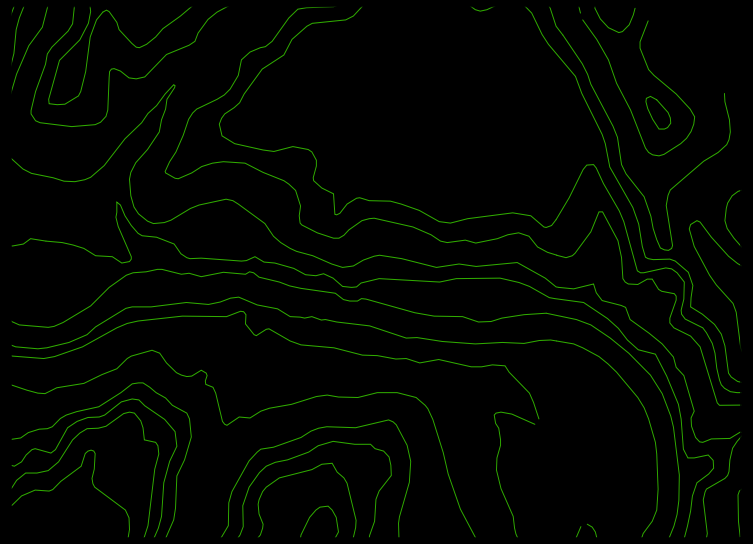




Kronhusbodarna, Göteborg

Vector, e.g. .shp

- Geographic features represented as:
  - Points
  - Lines
  - Polygons (areas)
- Features can have associated attributes

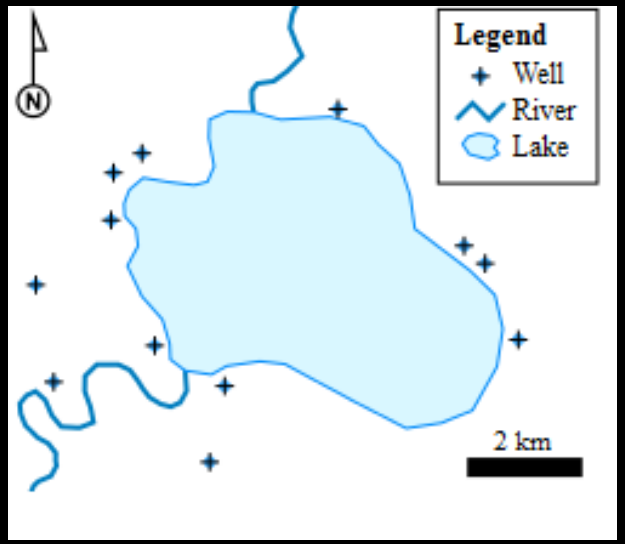


Contours (isolines)

Data formats

Shapefiles (.shp) – Vector

- One of the most widely supported GIS vector formats, developed by ESRI.
- The format has a publicly available specification, allowing other GIS software to read and write Shapefiles.
- Stores points, lines, or polygons, with associated attributes (e.g. name, temperature, population).
- Each Shapefile contains only one geometry type, but may also include 3D coordinates (Z).



Mandatory files :

.shp — shape format – the feature geometry itself
.shx — shape index format – a link between the .shp and the .dbf; allows fast seeking
.dbf — attribute format – columnar attributes for each shape

Example of optional files :

.prj — projection format – the coordinate system and projection information

Data formats

GeoPackage (.gpkg)

GeoPackage is an **open GIS data format standardized by OGC**, designed for storing and exchanging geospatial information.

A GeoPackage is based on an **SQLite database**, stored as a single .gpkg file.

A GeoPackage can contain:



- **Multiple vector layers** (points, lines, polygons) with attributes
- **Raster data** (primarily tiled imagery and raster maps)
- **Non-spatial tables** and metadata
- Additional functionality through extensions

Advantages:

- A single file can contain multiple GIS layers and tables.
- Data can be accessed and modified directly using **SQL**.
- Portable and platform-independent, with no database server required.
- Suitable for offline applications and environments with limited connectivity.

Data formats

```
<Placemark>
 <name>Gothenburg</name>
 <Point>
  <coordinates>11.97,57.71,0</coordinates>
 </Point>
</Placemark>
```

XML-based GIS formats: GML and KML

Extensible Markup Language (XML)

- A text-based markup language for storing and exchanging structured information in a machine-readable format.
- Uses tags to describe and organize data.

Geography Markup Language (.gml) — Primarily vector

- XML-based format for representing and exchanging geographic information.
- Supports geometries, attributes, coordinate reference systems, topology, and coverages.
- Developed and standardized by OGC, also adopted as an ISO standard.
- Primarily designed for describing geographic features and their properties.

Keyhole Markup Language (.kml)

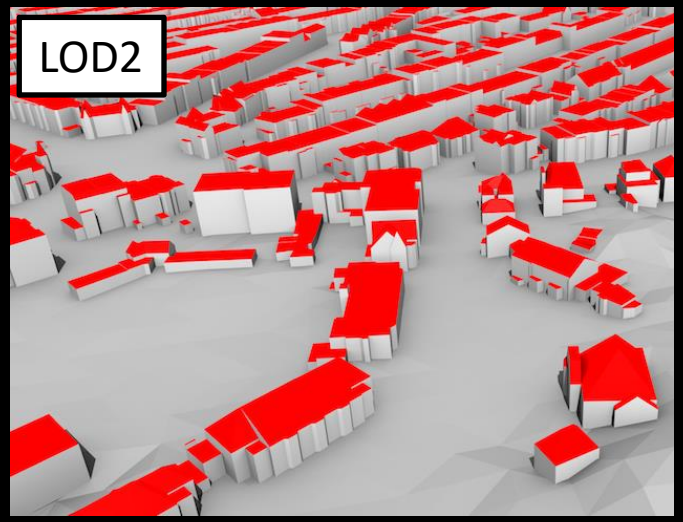
- XML-based format primarily designed for geographic visualization, particularly in Google Earth.
- Supports points, lines, polygons, labels, styling, and image overlays.
- Originally developed by Keyhole/Google; now an OGC standard.
- .kmz files are compressed ZIP archives containing KML and optionally other resources.

Data formats

CityGML – 3D City Models

CityGML is an OGC standard based on GML, developed for representing and exchanging 3D city models.

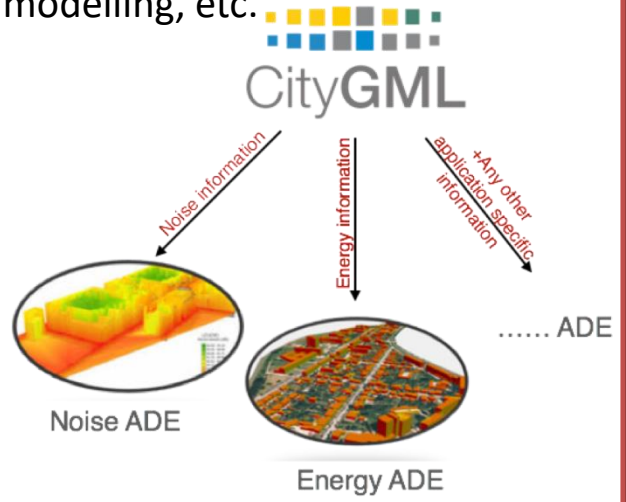
- Describes the geometry, attributes, and semantics of urban objects, such as buildings, roads, and vegetation.
- Supports different Levels of Detail (LOD0–LOD3), ranging from simplified building models to detailed 3D representations.
- Can also describe the appearance of objects, including colours and textures.





Application Domain Extensions (ADEs) for CityGML

An ADE is defined in an extra XSD (XML Schema Definition) file with its own namespace. ADEs can be defined by information communities which are interested in specific application fields. ADEs are increasingly being used in creating application specific extensions like for energy modelling, modelling topographic data, indoor modelling, noise modelling, etc.



Data formats

GPS exchange format (.gpx) – Vector

GPX is an open, XML-based format for storing and exchanging GPS data between devices and software applications.

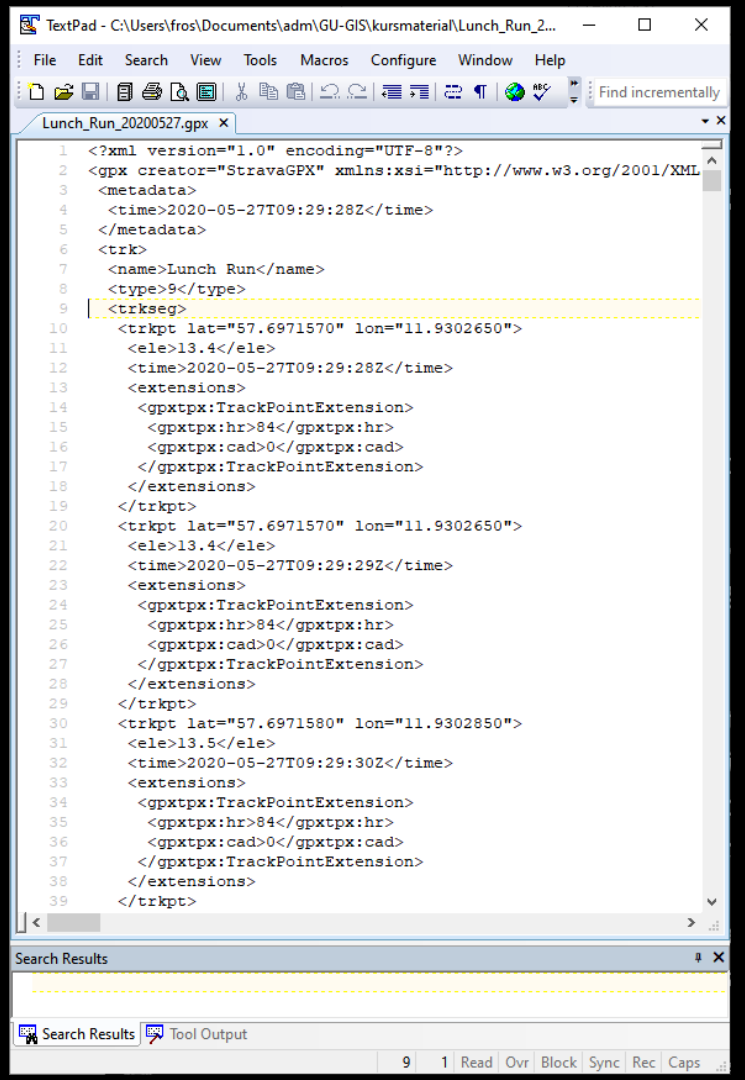
Originally developed by Topografix, it is commonly used for outdoor navigation, hiking, cycling and tracking movements.

GPX supports three basic feature types:

- Waypoints: Individual geographic locations (points).
- Routes: Sequences of points defining a planned path.
- Tracks: Recorded sequences of positions describing movement over time.

Other characteristics:

- Uses WGS 84 coordinates (latitude/longitude).
- Can store additional information, such as elevation, timestamps and names.
- Supported by QGIS and GDAL/OGR.

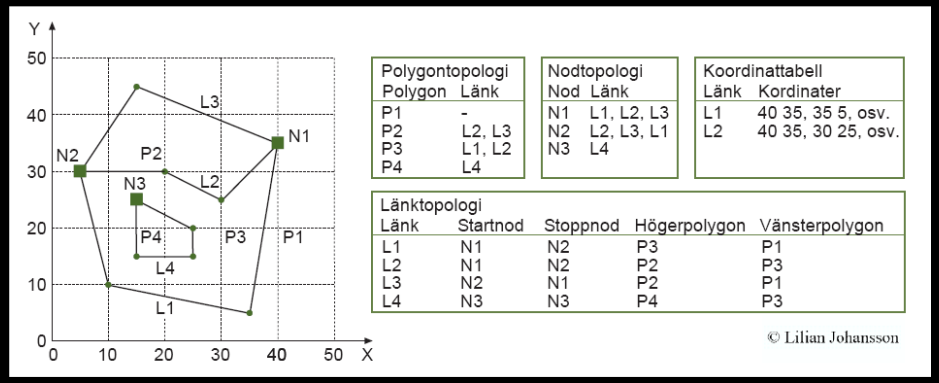


Data formats

The GRASS vector data model – Topological vector model

GRASS GIS uses a topological vector model, storing both geometries and their spatial relationships (e.g. connectivity and adjacency).

- A vector map can contain points, lines and areas.
- Areas are represented by connected boundaries, rather than independent closed polygons.
- Boundaries between adjacent areas are stored only once and shared by both areas.
- Boundaries must form closed rings without gaps to define valid areas.
- Areas are identified by centroids (label points), which link the areas to their attributes.
- Different geometry types can be stored within a single GRASS vector map.
- GRASS layers allow geometries to be associated with different categories and attribute tables.



Polygontopologi

Polygon	Länk
P1	-
P2	L2, L3
P3	L1, L2
P4	L4

Nodtopologi

Nod	Länk
N1	L1, L2, L3
N2	L2, L3, L1
N3	L4

Koordinattabell

Länk	Kordinater
L1	40 35, 35 5, osv.
L2	40 35, 30 25, osv.

Länktopologi

Länk	Startnod	Stoppnod	Högerpolygon	Vänsterpolygon
L1	N1	N2	P3	P1
L2	N1	N2	P2	P3
L3	N2	N1	P2	P1
L4	N3	N3	P4	P3

© Lilian Johansson

Data formats

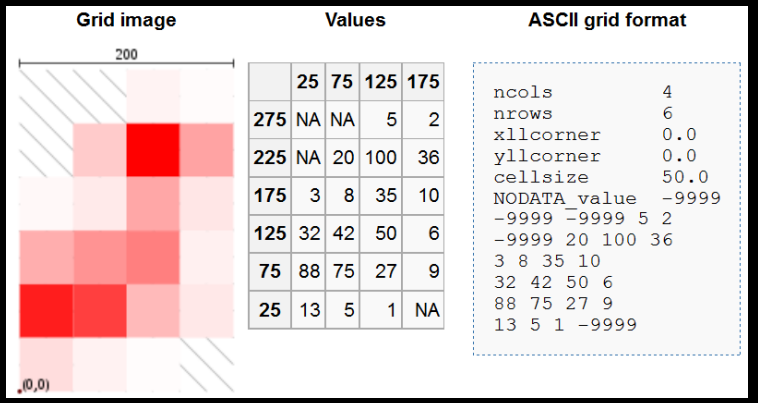
Esri grids – Raster

Esri Grid is a raster data format developed by **Esri**, commonly used for storing gridded data such as elevation, temperature and land cover.

Two related formats exist:

- Arc/Info Binary Grid (proprietary)
  - Stores raster data in a directory containing multiple binary files (.adf)
  - Developed for raster storage and analysis in Esri software
- Arc/Info ASCII Grid (open specification)
  - Stores raster data as plain text in a single file (.asc)
  - Contains a header describing the grid dimensions, geographic position and cell size, followed by the raster values
  - Easy to inspect and exchange between different GIS applications

Both formats can be read using GDAL



	25	75	125	175
275	NA	NA	5	2
225	NA	20	100	36
175	3	8	35	10
125	32	42	50	6
75	88	75	27	9
25	13	5	1	NA

```
ncols         4
nrows         6
xllcorner     0.0
yllcorner     0.0
cellsize      50.0
NODATA_value  -9999
-9999 -9999 5 2
-9999 20 100 36
3 8 35 10
32 42 50 6
88 75 27 9
13 5 1 -9999
```

Data formats

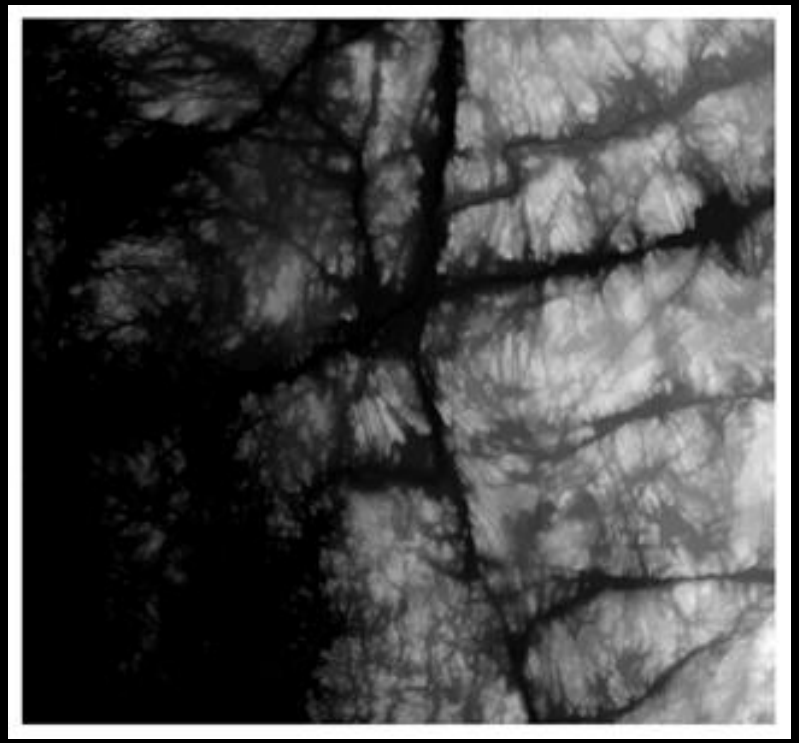
GeoTIFF and TIFF (.tif) – Raster

TIFF (Tagged Image File Format) is a general-purpose raster image format.

GeoTIFF is an extension of TIFF that includes geographic information directly in the file:

- Coordinate Reference System (CRS)
- Spatial position and pixel resolution
- One or multiple raster bands (e.g. elevation or RGB imagery)

GeoTIFF is an open format standardized by OGC and widely used in GIS.
A regular TIFF without embedded geographic information can be georeferenced using a separate world file (.tfw).



A DEM represented as a TIFF file.

JPEG (.jpg) — Raster

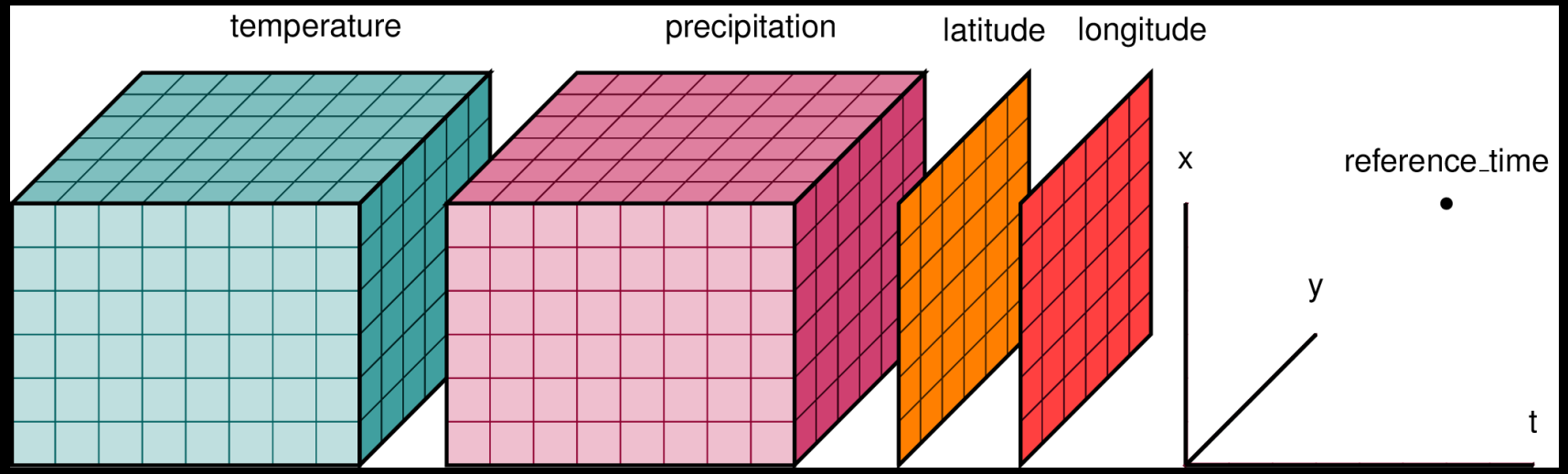
- Common image format using lossy compression.
- Can also be georeferenced using a world file (.jgw).
- Suitable for imagery, but generally not for numerical raster analysis where exact pixel values must be preserved.

Data formats

The Network Common Data Form (NetCDF)

NetCDF is an open, self-describing data format developed by Unidata/UCAR for storing and exchanging scientific data.

- Stores multidimensional arrays, e.g. 3D or 4D datasets with longitude, latitude, elevation/depth and time.
- Can contain multiple variables (e.g. temperature, pressure, precipitation) within the same file.
- Includes metadata describing variables, dimensions, units and other properties.
- Commonly used in meteorology, climatology and oceanography.
- Supported by GDAL and QGIS.



Data formats (web)

Web GIS Services — WMS and WMTS

OGC has developed several standards for accessing geospatial information over the internet, commonly referred to as **OGC Web Services (OWS)**.

- Allow GIS applications to access geographical information stored on remote servers.
- Different services provide different types of data (e.g. map images, vector features, raster data).
- Supported by GIS applications such as QGIS.

Web Map Service (WMS)

- Provides georeferenced map images over the internet (e.g. PNG, JPEG).
- The client specifies the geographical extent, CRS and layers to display.
- Maps are typically rendered on demand by a server, e.g. QGIS Server or GeoServer.
- Provides a visualization of spatial data, rather than direct access to the underlying geometries.

Web Map Tile Service (WMTS)

- Similar to WMS, but provides maps as predefined image tiles.
- Tiles can be cached for faster delivery, particularly useful for web maps.
- Uses predefined zoom levels and tile grids.

Data formats (web)

Web Feature Service (WFS)

WFS is an OGC standard providing direct access to geographic features (points, lines, polygons) and their attributes over the internet.

Unlike WMS (map images), WFS provides the actual vector geometries and attributes, allowing further GIS analysis.

Features are commonly transferred using GML, but other formats such as GeoJSON may also be supported.

Transactional WFS (WFS-T)

Some WFS servers also support editing operations (create, update, delete features)

WFS can be provided by servers such as GeoServer and QGIS Server.

Data formats

Web Coverage Service (WCS)

WCS is an OGC standard providing access to geospatial coverages over the internet.

- Coverages represent phenomena whose values vary across space and/or time (e.g. elevation, temperature, precipitation).
- Primarily used to access raster and multidimensional data.
- Unlike WMS, WCS provides the underlying data values, allowing further analysis.
- Data can be delivered in formats such as GeoTIFF or NetCDF, depending on server capabilities.

Web Processing Service (WPS)

WPS is an OGC standard that allows geospatial processing operations to be executed on a remote server.

- Clients can request GIS analyses without processing the data locally.
- Examples include buffer operations, polygon overlays, and NDVI calculations.
- The server performs the requested analysis and returns the results.
- The standard defines how processing operations are described, requested and executed, rather than which algorithms must be available.

Data formats

Choosing a Suitable GIS Data Format

Compatibility between GIS formats is generally good, largely thanks to libraries such as GDAL/OGR.

Data management

- How should the data be organized and stored?
- Single file, multiple files, or spatial database?
- Does the format support multiple layers and datasets?

Functionality

- Can the format store all the required information (geometries, attributes, CRS, raster bands, time dimensions, etc.)?
- Does it support the required operations, such as editing and updating?

Interoperability

- Is the format supported by the software used by you and others?
- Can data be exchanged without losing important information?

Performance and efficiency

- File size, compression and read/write performance.
- Is the format suitable for large datasets or accessing data remotely?

European Petroleum Survey Group (EPSG)

Coordinate Reference Systems — EPSG

The EPSG Geodetic Parameter Dataset contains standardized definitions of coordinate reference systems (CRS), coordinate transformations and related geodetic parameters.

Originally developed by the European Petroleum Survey Group, it is now maintained by IOGP.

- Coordinate reference systems are identified by unique EPSG codes.
- Definitions include information about the datum, ellipsoid, coordinate axes, units and, where applicable, map projection parameters.
- GIS applications use these definitions to interpret spatial coordinates and transform data between coordinate reference systems.

Examples:

- EPSG:4326 — WGS 84 (geographic coordinates, degrees)
- EPSG:3007 — SWEREF 99 12 00 (projected coordinates, metres)
- EPSG:3006 — SWEREF 99 TM (Swedish national projection, metres)

CRS definitions can also be expressed using Well-Known Text (WKT), for example in a Shapefile's .prj file.

```
GEOGCS["GCS_WGS_1984",
  DATUM["D_WGS_1984",
    SPHEROID["WGS_1984",6378137,298.257223563]],
  PRIMEM["Greenwich",0],
  UNIT["Degree",0.0174532925199433295]]
```

PROJ (proj.org)

PROJ — Coordinate Transformations

PROJ (proj.org) is an open-source library for transforming geospatial coordinates between different Coordinate Reference Systems (CRS).

PROJ supports:

- Map projections: Geographic coordinates (latitude/longitude) to projected coordinates (metres).
- Inverse projections: Projected coordinates back to geographic coordinates.
- Datum transformations: Transformations between different geodetic reference systems.

For example, transforming coordinates from:

EPSG:4326 (WGS 84) — Geographic coordinates (degrees)
to
EPSG:3007 (SWEREF 99 12 00) — Projected coordinates (metres)

PROJ is used by many GIS applications and libraries, including QGIS, GDAL, GRASS GIS and PostGIS.

Geodetic datum
A reference framework that defines how a coordinate system is positioned, oriented and scaled relative to the Earth.
It establishes the relationship between coordinates and physical locations on the Earth's surface.

Open Geodata

OpenStreetMap (OSM)

OpenStreetMap is a community-driven initiative to create and provide free and open geographic data to anyone.

- A collaborative project where volunteers collect, edit and maintain geographic information worldwide.
- Contributors use GPS recordings, aerial imagery, field surveys and local knowledge.
- Contains information about roads, buildings, land use, points of interest and much more.
- The OpenStreetMap Foundation (OSMF) is a non-profit organization supporting the project, its infrastructure and development.

Data format and licensing

- Data can be downloaded in the XML-based OpenStreetMap format (.osm).
- Data is licensed under the Open Database License (ODbL 1.0), requiring attribution and, in certain cases, share-alike distribution.

Website: https://www.openstreetmap.org/

Open Geodata

Natural Earth Project

Natural Earth is a public-domain geographic dataset providing free vector and raster data, primarily for global and regional mapping.

- Available at three map scales:
  - 1:10 million – Most detailed
  - 1:50 million – Intermediate detail
  - 1:110 million – Least detailed

- Contains both physical data (coastlines, rivers, lakes, terrain) and cultural data (countries, administrative boundaries, cities).
- Vector data is available in formats such as Shapefile and GeoPackage, and raster maps in TIFF format.
- Designed primarily for cartographic visualization, rather than detailed local GIS analysis.
- All datasets are in the public domain (no attribution required).

https://www.naturalearthdata.com/



Open Geodata

Other Sources of Geodata

Swedish geodata

- Saccess (Lantmäteriet) — Historical satellite imagery covering Sweden from the 1970s onwards.
- Geodata Extraction Tool (GET) — Swedish geodata available to universities for education and research, including data from Lantmäteriet, SGU and SCB.
  https://maps.slu.se/get/
- Geodataportalen — National catalogue for discovering geodata and services from Swedish authorities.
  https://www.geodata.se/
- Lantmäteriet Geotorget — Access to Swedish maps, elevation models, aerial imagery and other geodata.
  https://geotorget.lantmateriet.se/
- Municipalities — Many Swedish municipalities provide local geodata through open-data portals.

International geodata

- Copernicus (EU) — Satellite imagery and environmental data, including Sentinel missions.
  https://dataspace.copernicus.eu/
- ESA Climate Change Initiative — Global satellite-derived climate datasets.
  https://climate.esa.int/en/odp/
- USGS / NASA — Landsat imagery, global elevation models (e.g. ASTER GDEM) and other Earth observation data.
  https://earthexplorer.usgs.gov/

Abbreviations and acronyms - concepts

Abbreviations and acronyms - concepts

Abbreviations and Acronyms — Libraries and Concepts

GDAL: Geospatial Data Abstraction Library — for accessing and processing geospatial data.

OGR: Historical name for GDAL's vector data functionality (OGR Simple Features Library).

PROJ: Open-source library for coordinate transformations.

OGC: Open Geospatial Consortium — develops open geospatial standards.

OSGeo: Open Source Geospatial Foundation — supports open-source geospatial software projects.

CRS: Coordinate Reference System — defines how coordinates relate to locations on Earth.

EPSG: Geodetic parameter dataset containing CRS definitions, identifiers and coordinate operations.

WKT: Well-Known Text — textual representation of geometric objects or coordinate reference systems.

Datum: Reference framework defining how a coordinate system is anchored to the Earth.

Topology: Spatial relationships between features, such as connectivity, adjacency and containment.

Abbreviations and acronyms - concepts

Abbreviations and Acronyms — Formats and Web Services

Data formats

- XML: Extensible Markup Language — structured text-based data representation.
- GML (.gml): Geography Markup Language — geographic information, primarily vector features.
- KML (.kml): Keyhole Markup Language — geographic visualization.
- GPX (.gpx): GPS Exchange Format — waypoints, routes and tracks.
- GeoPackage (.gpkg): SQLite-based geospatial data format.
- GeoTIFF (.tif): Raster format with embedded geographic referencing information.
- NetCDF (.nc): Network Common Data Form — multidimensional scientific data.

OGC Web Services

- WMS: Web Map Service — map images.
- WMTS: Web Map Tile Service — map image tiles.
- WFS: Web Feature Service — vector features and attributes.
- WCS: Web Coverage Service — raster and coverage data.
- WPS: Web Processing Service — remote geospatial processing.